

Supplementary Material: Task-Relevant Information Acquisition for Ultrasonic Inspection of Impacted Composites

Author information withheld for review

1 Purpose and statistical units

This supplement exposes the audit-heavy records behind the main manuscript. It does not introduce a second method, new training, or additional performance claims. Physical specimens are the first statistical unit and the six held-out experimental domains are the second. Repeated state, checkpoint, action, and learner-pair rows are computational records rather than independent samples. All oracle, teacher, and component-substitution records are retrospective and non-deployable unless a file explicitly marks a row as deployable.

2 Machine-readable evidence index

Files	Content
S01–S02	Full domain-by-domain scalar/spatial and static-observability metrics.
S03–S04	Full checkpoint and domain records for strict-OOF conditional-value evolution.
S05–S07	Complete real, position-only, shuffled, static, reconstruction, and full-field control records.
S08–S09	Dynamic-valuation regret, Spearman, NDCG, recall, utility, and domain records.
S10–S11	Valuation/planning component substitutions and their domain-level effects.
S12	Current greedy, beam, lookahead, and retrospective reachable-pool planning records by domain.
S13–S14	Learner fits and all specimen-level learner-pair rank, top- k , best-action, and spatial metrics.
S15–S16	Feedback-effect strata and association diagnostics for tested mechanism variables.
S17	Paper source paths and SHA-256 provenance authority.
S18	External-dataset feasibility audit; no external method-performance evaluation is present.

3 Part I: Information characterization details

S01 provides all domain-level full-field candidate metrics, including the matched metadata-plus-surface B-family path and the distinct independent metadata-only I-family estimator. The registered confirmatory claim uses the B-family scalar-to-spatial contrast. S12–S14 preserve the full

planning and learner-sensitivity records used to show that measurement value is conditioned on the downstream predictor and decision set rather than being an intrinsic property of a location.

4 Part II: State-conditioned valuation and source controls

S02 contains the full static-observability comparison. S03–S04 contain every tested checkpoint transition and value source for the conditional teacher. S05–S07 expose all state-matched content controls, including the acquired-position/history and registered normalized-RGB-MSE reconstruction comparisons. The positions-only input inherits the frozen acquisition history and is not a pure geometry-only control. S08–S09 retain secondary rank metrics and the shuffled-content boundary. These records delimit specimen-specific content attribution for the tested legal state.

5 Part II: Decision realization and deployment calibration

S10–S11 expose the full component-substitution matrix and domain effects. S12 contains the bounded two-action reachable-pool diagnostics; the retrospective near-oracle is not a deployable policy and does not establish arbitrary-horizon planning regret. S15–S16 retain the full feedback strata and association analyses. The frozen stress test retains the no-feedback reference, and the learned endpoint remains not performance-superior to the strongest deployable reference.

6 Provenance and external-data boundary

S17 binds the paper-specific canonical authority to the immutable source artifacts. ‘SUPPLEMENTARY_DATA_MANIFEST.csv’ independently binds each copied supplement file to its source digest. S18 is an audit of external dataset feasibility only. It contains no external replication, scanner-time validation, or industrial deployment result.